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Distribution and Characteristics of Blackwater Rivers and Streams of the Contiguous United States

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Key Points:

- There are a substantial number of blackwater rivers in multiple ecoregions of the United States
- The water chemistry of blackwater rivers and streams is often very different than that of non-blackwater rivers and streams
- We explore these differences and identify research needs that would contribute to their management and protection

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract

Blackwater rivers and streams have stained or tea-colored water from tannins released by decaying plant matter. Natural conditions in these waters often differ from non-blackwater systems. For example, the pH and oxygen levels in waters can be very low, but completely natural. We examined an existing USEPA data set and found that blackwaters existed across the contiguous United States but were most common in the east. Water chemistry showed differences between blackwater and non-blackwater sites, but differences were not consistent across ecoregions making national scale generalizations difficult. Physical habitat data analysis did not show dramatic differences between blackwater and non-blackwater sites. Blackwater typically arises from streams that drain tannin-rich bogs/muskeg and wetlands, so as expected blackwater sites had a shorter Euclidean distance to wetlands than non-blackwater sites and existed in watersheds with a higher percentage of wetland habitat. Blackwaters in Northern and Temperate Plains tended to have higher acid neutralizing capacity, conductivity, and lower True Color; a visual color scale used for water purity. We posit that differences were because Color and Dissolved Organic Carbon at these sites were from buried wetland deposits rather than contemporary wetland habitats. Research needs that may increase our understanding and management of blackwaters include development of an operational definition that includes a classification framework and reference conditions for different blackwater types, identification of stressors and their associated dynamics that negatively impact blackwater systems, and development of data-driven, consistent, and repeatable assessment methods, including development of targets, protective of unique conditions in blackwater rivers and streams.

Plain Language Summary

Blackwater rivers and stream systems have stained or tea-colored water due to tannins released by decaying plant matter. Although blackwaters exist across the contiguous United States, they are most common in the eastern United States. Beyond their distinct appearance, the chemistry of blackwater systems can be very different from other stream types. For example, the pH and amount of oxygen in the water can be very low, but completely natural. As a result of these and other characteristics, the fish, plants, and animals that live in blackwaters can also be unique. Because of these considerations, management of these systems can be challenging. To address these challenges, U.S. Environmental Protection Agency scientists are studying these systems. One of their early findings is that different types of blackwaters exist across the country. The team has highlighted the need for a classification system to distinguish among these types. Once this system exists, scientists can begin to identify what is required to protect blackwater systems and the services they supply.

1. Introduction

The term "blackwater" is commonly used to describe low-gradient rivers and streams with characteristically tea-colored waters due to high levels of tannins from decaying vegetation. Other terms used globally to describe these systems include "tannic" (Fraley et al., 2018; Largier, 1986), "brownwater" (Benedetti et al., 2006; Clair & Freedman, 1986; Lotzkes et al., 2021; Stahle et al., 1985), "humic" (Heikkinen, 1994), and "stained" (McKinley, 2013; Waldo, 2019). Blackwater rivers and streams exist around the world with substantial numbers existing in Africa (Anwana et al., 2021; Thieme et al., 2005), Asia (Alkhatib et al., 2007; Lotzkes et al., 2021),

10.1029/2023WR035529

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Europe (Grzybkowska et al., 1996; Heikkinen, 1994), North America (Benke & Wallace, 2015; Meyer, 1990; Zampella et al., 2007, 2008), and South America (Duncan & Fernandes, 2010; Ríos-Villamizar et al., 2013; Vegas-Vilarrubia et al., 1988). In the preceding papers, blackwater rivers and streams are characterized by clear (i.e., low turbidity) or stained watercolor; water chemistry that includes low pH/acidic, low dissolved oxygen, high dissolved organic carbon (DOC), high tannins (i.e., humic and fulvic acids), low conductivity, and low nutrients; systems that have a low gradient and a sandy bottom; and unique biota. Although these characteristics are widely used to describe blackwater rivers and streams, blackwater systems not fitting this paradigm also exist (Flotemersch, 2023; Smock & Gilinsky, 1992). For example, the blackwater rivers of Blackwater Falls State Park, West Virginia (www.wvstateparks.com), and "The Root Beer Falls" of Tahquamenon State Park in Michigan (https://www.michigan.org) are clearly not low gradient, nor dominated by a sandy bottom. Therefore, examination of these descriptive characteristics is warranted.

The United States Clean Water Act (CWA Section 305(b)) directs the United States Environmental Protection Agency (USEPA) and states, territories, and other jurisdictions to report on the condition of waters in the U.S., which includes blackwater rivers and streams. Condition assessments of aquatic habitats in general are accomplished based on physical, chemical, and biological data collected at sample sites. Data from sites are then compared to scientifically determined values that represent the bounds of acceptable conditions. Often, these values are set based on conditions at least-disturbed sites (i.e., the best available sites (Stoddard et al., 2006; USEPA-WQS, 2023)). However, in some blackwater streams, natural conditions may still exceed legally established thresholds, which are largely based on non-blackwater sites. This is a regular occurrence when regulatory organizations attempt to assess the condition of blackwater rivers and streams (Flotemersch, 2023).

Blackwater rivers and streams exist in various locations across the United States. Those in the Atlantic coastal plains are probably the most well-known and well-studied. The relatively undisturbed condition of blackwater systems in southern parts of the coastal plains resulted in blackwater rivers and streams in this area being the focus of a flurry of stream ecology research efforts in the late 1970s that continues through to today (Benke & Wallace, 2015; Colvin et al., 2020; Meyer, 1990; Smock & Gilinsky, 1992). Considerable research has also been conducted in the northern extent of the Atlantic coastal plains to better understand and conserve blackwater habitats in the New Jersey Pinelands National Reserve (NJPC, 2022). Several state agencies in this region have developed approaches for identifying, assessing, and reporting condition of blackwater habitats. However, these approaches vary widely in both scope and state of development (Flotemersch, 2023).

Despite the above-mentioned research efforts, many questions remain about blackwater rivers and streams. How do they vary regionally? What factors drive these regional differences? How have biotic communities adapted to these unique systems? Are the general descriptive characteristics used in the literature to describe the physical habitat of blackwater systems accurate? And what management actions are required to adequately manage these systems in a way that sustains their existence? Our goal is to provide a first look at the distributional patterns and the chemical and physical characteristics of blackwater rivers and streams (hereinafter "blackwaters"), and their relationship to wetlands across the contiguous United States (CONUS) using data collected as part of the National Rivers and Streams Assessment (NRSA) surveys (USEPA-NRSA, 2022). The NRSA data set was selected for this study as it is a probability-based survey of rivers and streams that collects and analyzes physical and chemical data using standardized field and laboratory methods. Our specific objectives were to (a) identify the general distribution of blackwaters in the CONUS, (b) estimate the extent of the blackwater rivers and streams, (c) describe the chemical and physical habitat characteristics of blackwaters in the CONUS and within ecoregions and compare findings with how blackwaters are generally described in the existing literature, (d) explore the relationship of blackwaters with wetlands, and (e) provide a list of the next steps in research needed to support improved understanding and protection of blackwater rivers and streams. Meeting these objectives will contribute significantly to the goal of developing a data-driven set of consistent and repeatable condition screening criteria for blackwaters—an essential step in advancing the protection of this under-recognized aquatic habitat.

2. Methods

2.1. Study Area

The study area for this effort is the CONUS (Figure 1). This area was further divided into the nine aggregated Omernik ecoregions (hereinafter "ecoregions") developed by Herlihy et al. (2008). Each of these are aggregations of Level III ecoregions (Omernik, 1987; Omernik & Griffith, 2014), the boundaries of which are defined so that

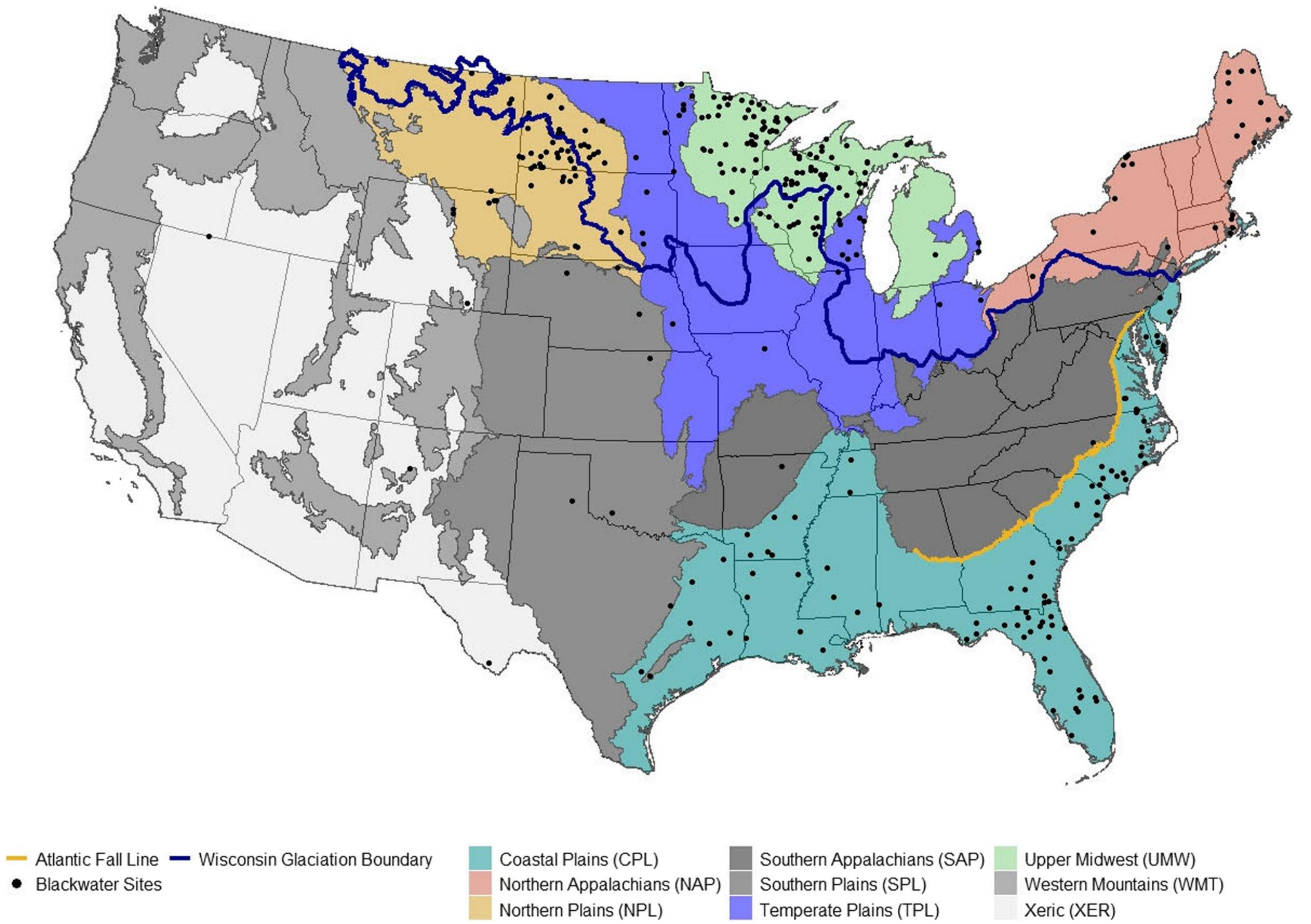

Figure 1. The National Rivers and Streams Assessment (probability and handpicked) sites identified as blackwater systems (i.e., Color ≥50 PCU and DOC ≥ 10 mg L^{-1}), their distribution within the nine aggregated ecoregions and in relation to the Atlantic Fall Line (yellow line) and Wisconsin glaciation (blue line). White and gray ecoregions were not included in data analysis due to insufficient number of sites that met our criteria for blackwater.

each ecoregion contains a unique set of environmental characteristics such as climate, vegetation, soil type, and geology. The USEPA uses these nine ecoregions in the National Aquatic Resource Surveys (NARS) program (USEPA-NARS, 2022) to account for geoclimatic factors that influence chemical and physical habitat and affect the quality, health, and integrity of ecosystems (Olsen, 2007; Omernik, 2004).

2.2. Data Set

The initial data set for this study is a composite of data collected from sites sampled as part of NRSA surveys conducted in 2008–2009, 2013–2014, and 2018–2019 (USEPA-NRSA, 2022). The NRSA surveys use a probability-based survey design (Olsen & Peck, 2008; Olsen et al., 2012) that ensures that the survey samples a balance of river and stream size classes. The spatial distribution of sampled sites is also controlled to assure distribution across the contiguous United States. At each site, physical, chemical, and biological data are collected at rivers and streams that meet the target definition using standardized field and laboratory methods. The definition of the target population for NRSA is rivers and streams within the 48 contiguous states that have flowing water during the summer low-flow season that roughly spans from June 1st through September 30th. The goals of the NRSA are to estimate the physio-chemical and biological condition of rivers and streams and to identify important stressors negatively affecting their condition. An additional longer-term goal is to analyze long-term trends of condition and the actions needed to assure long-term protection.

Both probability-based and handpicked sites are included in the NRSA. Handpicked sites are individually identified (i.e., they do not follow the survey design) and are sampled because they are expected to meet

Table 1
Chemical Analyses and Physical Habitat Descriptors Included in the Analysis

Parameter type	Analyte/Assessed feature (Abbreviation)	Units
Chemical	Dissolved Organic Carbon (DOC)	mg/L
	True Color (Color)	PCU
	Conductivity (COND)	µS/cm
	Turbidity (TURB)	NTU
	Total Phosphorus (Total P)	µg/L
	Total Nitrogen (Total N)	µg/L
	pH	–
	Acid-Naturalizing Capacity (ANC)	µeq/L
Channel Characteristics	Elevation	m
	Mean water surface gradient (% Slope)	%
	Mean thalweg depth -wetted value (Depth)	Cm
	Mean wetted width (Width)	M
	Mean wetted width × thalweg depth Width × Depth)	m^2
	Incised height—water surface to first depositional terrace (Incised height)	m
Riparian Characterization	Riparian cover—summed over canopy, understory, and ground cover layers (Riparian Cover)	Proportion
	Riparian woody cover—summed over canopy, understory, and ground cover layers (Riparian woody cover)	Proportion
	Coniferous riparian canopy presence (Coniferous canopy)	Proportion
	Deciduous riparian canopy presence (Deciduous canopy)	Proportion
	Riparian ground cover—Barren or Duff (Grd Cvr Barren)	Proportion
	Riparian ground cover—non-woody vegetation (Non-wdy Grd Cvr)	Proportion
	Riparian ground cover—woody vegetation (Woody Grd Cvr)	Proportion
Streambed surface Substrate Characterization	Log_{10}(Dgm) = Log_{10}(geometric mean substrate diameter) (Substrate Characterization)	Log_{10} (mm)
	Substrate, percent fines (silt/clay; <0.06 mm) (% Fines)	%
	Substrate, percent sand (0.06–2.0 mm) (% Sand)	%

least-disturbed (i.e., "reference") criteria (Stoddard et al., 2006; Whittier et al., 2007) and, therefore, may be potentially useful in the establishment of benchmarks for the evaluation of probability sites (USEPA, 2020). A limitation of this data set is that the targeted sampling season does not support examination of seasonal variability in blackwater rivers and streams. Data from Alaska and Hawaii were not included in the analysis because the field and laboratory methods used in those states are not fully consistent with those used in the 48 contiguous states. Field protocols used in the surveys are detailed in USEPA (2009, 2013a, 2013b). Those relevant to this study are summarized below.

For the collection of all variables, a sample reach was established around the sample point (i.e., specific coordinates) sufficient for characterization of the physical habitat and the instream biota (Hughes & Peck, 2008; Reynolds et al., 2003). For water quality variables (Table 1), one water grab sample was collected from the middle of the site for wadeable streams, and at the downstream end of the site for boatable rivers (USEPA, 2009). Iced samples were then shipped overnight to analytical laboratories. In the laboratory, values for pH and conductivity (COND) were measured using meters. Sulfate and chloride concentrations were measured by ion chromatography. Total phosphorus (Total P) and total nitrogen (Total N) were measured by acid persulfate digestion and colorimetry using a flow injection analyzer. Dissolved organic carbon (DOC) was measured using a UV/persulfate carbon analyzer. Acid-neutralizing capacity (ANC) was measured using an automated acidimetric titration to pH < 3.5, with modified Gran plot analysis. True Color (Color) was determined with a visual comparator, and turbidity (TURB) was measured with a nephelometer. Lab methodologies are detailed in USEPA (2012).

Channel and riparian physical habitat variables (Table 1), including channel dimensions, substrate, instream habitat features, and riparian vegetation cover, were collected as described in Kaufmann et al. (1999, 2022),

10.1029/2023WR035529

Hughes and Peck (2008), and USEPA (2009, 2013a, 2013b). Riparian vegetation cover refers to woody and herbaceous cover of trees, shrubs and ground along the sample stream or river channel. Instream habitat features include large and small wood, overhanging banks, inundated live trees, boulders, and rock ledges. The areal cover of these riparian and instream features are measured as part of NRSA surveys because they moderate the input and transport of heat, water, sediment, nutrients and contaminants; and they provide habitat and refuge for instream organisms and at the land-water interface. Physical habitat measures were considered in the current study to help characterize rivers and streams identified as blackwater and to compare their characteristics to the those published elsewhere. Multiple measurements were made along the channel thalweg, and in instream and riparian plots at 11 evenly spaced transects along the sample reach. Woody riparian vegetation cover, substrate composition, and wetted width and depth data were collected at each transect through use of standardized field forms. Between transects, crews determined water surface slope and collected depth, width, substrate, and habitat type (e.g., pool, glide, riffle) data at systematic intervals. Field data were converted into physical habitat metrics (Table 1) following the methodology described in Kaufmann et al. (1999, 2022). In brief, riparian cover metrics are summarized from standardized measurements at both banks at each transect. Substrate data are based on visual estimation of the size of 105 bed surface particles (pebble count adapted from Wolman (1954)) in wadeable sites, and through use of a sounding rod in non-wadeable sites (Collins & Flotemersch, 2014). Bed surface percent variables are areal cover percentages and substrate size is the geometric mean diameter of the pebble counts (Kaufmann et al., 2022).

A subset of sites in the composite data set were sampled more than once (i.e., data collected from the same stream or river reach within/across years). Where this occurred, we only used data collected during the first visit to a site to prevent double counting and data bias. Although not the focus of this study, this pool of resampled sites likely offers information on temporal variability of blackwater sites within the summer low-flow sampling season used by the NRSA. We identified a subset of sites with Color ≥50 PCU and DOC ≥10 mg L^{-1} as likely blackwater sites. These two chemical analytes were selected because they are common attributes mentioned in the literature describing blackwaters in the United States (e.g., Benke & Wallace, 2015; Colvin et al., 2020; Flotemersch, 2023; Meyer, 1990; Smock & Gilinsky, 1992). The stated value DOC was selected based on ranges observed in the literature (e.g., DOC 4–74 mg L^{-1}) for sites described as blackwaters (e.g., Benke & Parsons, 1990; Meyer, 1986; Smock & Gilinsky, 1992). The blackwater sites selection criteria used in this study are not proposed as criteria for defining blackwater rivers and streams in the CONUS because these values would exclude a large number of sites with DOC or Color below these values. Again, we were targeting sites that were well within the range of what most would consider blackwater. All sites meeting selection criteria were plotted within the framework of the ecoregions to visualize distribution (Figure 1). Any ecoregion having <10 sites was excluded from ecoregional comparisons to assure sufficient site numbers existed in retained ecoregions to support robust exploration of ecoregional differences.

2.3. Estimating the Extent of Blackwater Rivers and Streams

The extent (percent and kilometers) of blackwater rivers and streams in the CONUS meeting the criteria used in this study (i.e., Color ≥50 PCU and DOC ≥10 mg L^{-1}) was estimated. Only probability-based perennial sites within the NRSA database were used for these estimates; handpicked sites were excluded (e.g., Olsen & Peck, 2008). Each probability site has a weighting factor (km) that is used for the accurate extrapolation of information provided by a set of sampled sites to larger subpopulations such as (a) analogous river and stream size categories, (b) ecoregions, and (c) to the full extent of all river and stream length across the CONUS that met the target definition (USEPA, 2020; Valliant et al., 2013; Zieliński et al., 2018). Extent estimates of blackwaters were calculated using the spsurvey R package (v. 5.3.0, Dumelle et al., 2022). Results are presented by ecoregion as percent stream length and estimated stream length (km).

2.4. Data Analysis

Descriptive and inferential statistics were used to compare the chemical and physical habitat characteristics of the blackwater to that of non-blackwater sites at the scale of the CONUS and within and across ecoregions. All analyses were run using R software (v. 4.2.1, R Core Team, 2022).

10.1029/2023WR035529

2.4.1. Water Chemistry and Physical Habitat

For water chemistry, box plots comparing blackwater and non-blackwater sites by ecoregion were generated. A Principal Components Analysis (PCA) of water chemistry data was also conducted to further explore blackwater site characteristics. Prior to the PCA, we checked for strong correlations among analytes. If strong correlations were identified (Spearman $r > 0.80$), only one of the correlated analytes was retained for the analysis. Among correlated variables, we opted to retain variables more easily measured in the field (e.g., conductivity over ANC) as these would be more relevant and useful to field crews needing to make field-based decisions regarding sampling strategies. Input variables were also examined to determine if distributions were highly skewed. To normalize the data distributions, a $\log_{10}(x)$ transformation was applied to COND, NTL, PTL, TURB, and DOC, as there were no 0 values. For Color, we applied a $\log_{10}(x + 1)$ transformation. No transformation was applied to pH. The PCA was run on a correlation matrix of the data (i.e., variables were centered and scaled) using the *prcomp* function in the *stats* R package (R Core Team, 2022). The first two principal components, along with variable loadings, were then plotted by ecoregion to identify patterns in the data.

This same analysis plan was followed for physical habitat data with the exception that highly skewed variables ranging from 0 to 1 were $\log_{10}(x + 0.1)$ transformed, and those ranging from 0 to 100 were $\log_{10}(x + 1)$ transformed.

2.4.2. Wetlands

Because of the importance of bogs and wetlands as sources of Color and DOC, we compared the Euclidean distance from sample points to the nearest wetland between blackwater and non-blackwater sites and among ecoregions. Euclidean distance was chosen instead of flow path distance because flow path can be difficult to measure in low-gradient systems (e.g., Ralph et al., 2021) and due to the complex multi-channel nature of wetlands (Liu & Wang, 2017). Furthermore, contributions to streams from isolated wetlands can be significant (e.g., Alexander et al., 2018; Leibowitz, Wigington, et al., 2018) and are underrepresented in flow path analysis. The Euclidean distance was calculated by generating a gridded raster (matrix of cells) and calculating the distance from each NRSA site to the closest cell containing a National Wetlands Inventory (NWI; www.fws.gov) wetland. Distance was measured from the center of the 30 m raster cell to the edge of the wetland.

To explore the relationship between wetlands and blackwaters, we extracted data from USEPA's Stream-Catchment (StreamCat) database to create box plots showing the percentage of land in the watershed classified as wetlands by the NLCD (NLCD 2011 class 90). StreamCat is an extensive database of landscape metrics for ~2.65 million stream segments and their associated catchments within the contiguous United States (StreamCat, 2022). A detailed description of the StreamCat Data set and its development is available (Hill et al., 2016). Within StreamCat, a watershed represents a set of hydrologically connected catchments, including all upstream catchments that flow to any local catchment. All calculations used the sample site location to identify the watershed. We viewed woody and herbaceous wetlands independently and then summed these to represent the total percentage of each watershed classified as wetland land cover.

3. Results

The composite data set contained data from 6,111 site visits. Removal of revisit/resample sites reduced this number to 4,389. Application of the screening criteria to identify sites with a high likelihood of being blackwater (Color ≥50 PCU and DOC ≥10 mg L^{-1}) resulted in a data set of 326 blackwater and 4,063 non-blackwater sites. Ecoregions with the most blackwater sites in the NRSA data set were in the Upper Midwest (UMW; $n = 106$ of 296) and Coastal Plains (CPL; $n = 95$ of 502). Substantial numbers also occurred in the Northern Plains (NPL; $n = 58$ of 313), Temperate Plains (TPL; $n = 27$ of 487), and Northern Appalachians (NAP; $n = 24$ of 480). Hand-picked blackwater sites were among these (i.e., UMW, $n = 27$; CPL, $n = 14$; NPL, $n = 3$; TPL, $n = 3$; NAP, $n = 1$). Blackwater sites were also identified in the Southern Plains (SPL; $n = 7$ of 383), Xeric (XER; $n = 4$ of 444), Southern Appalachians (SAP; $n = 3$ of 637), and Western Mountains (WMT; $n = 2$ of 521) ecoregions (Figure 1). Given the low numbers of blackwater sites in these last four ecoregions, they were not included in the analysis. This resulted in a total of 310 blackwater sample sites (probability and handpicked) across five ecoregions.

Table 2
Percent of Total Stream Estimated Length (km) and Estimated Length of Blackwater Streams in Five Aggregated Ecoregions

	Criteria: Color ≥50 PCU and DOC ≥ 10 mg L^{-1}		All streams
Ecoregion	% Length (SE)	Length km (SE)	Total estimated length km (SE)
Coastal Plains (CPL)	18.2(2.5)	76,565(11,132)	421,284(23,478)
Northern Appalachians (NAP)	13.6(2.7)	26,073(5,711)	191,741(9,229)
Northern Plains (NPL)	13.5(1.7)	6,633(866)	49,149(1,346)
Temperate Plains (TPL)	7.6(1.9)	29,783(7,730)	390,247(23,692)
Upper Midwest (UMW)	25.0(2.8)	32,691(3,392)	131,346(8,737)

Note. Km = kilometer, SE = Standard Error.

The distributional extent of many of the sites was bounded by two major geomorphic features (Figure 1)—one in the southeastern CONUS and one in the northern CONUS. In the southeastern CONUS, blackwater sites were concentrated on the Atlantic side of the Atlantic Fall Line where the Piedmont and Atlantic coastal plains meet (Freitag et al., 2012). This demarcation aligns with the border between the CPL and SAP ecoregions. The topography east of the Fall Line largely consists of flat plains with numerous wetlands, including the Florida Everglades and Okefenokee Swamp (USEPA-Ecoregions, 2022). The occurrence of blackwater rivers in streams in this area is well documented (Benke & Wallace, 2015; Colvin et al., 2020; Flotemersch, 2023; Meyer, 1990; Smock & Gilinsky, 1992). In the northern CONUS, sites were largely concentrated north of the southern extent of the Wisconsin Glaciation. This boundary also aligns with multiple ecoregional borders in that part of the CONUS. Wetland abundance is high north of this glacial boundary in response to glacial scouring that created depressions and sedimentary deposits (Fergus et al., 2017).

3.1. Estimating the Extent of Blackwater Rivers and Streams

The 239 NRSA sites we designated as blackwaters are a subset of the total number (1952) of CONUS probability survey sample sites in the five ecoregions that were the focus of analysis. We used this subset and its population expansion weights to quantify the extent of blackwaters in the study ecoregions (Note that 71 handpicked sites were excluded). In terms of total length of blackwater streams, the Temperate Plains had the smallest proportion of blackwater streams, at about 7.6%. Whereas the Upper Midwest had the greatest proportion of blackwater stream length (25%), the larger Coastal Plain had the greatest estimated blackwater stream length, 76,565 km—more than twice as much as in the UMW, TPL, or NAP, and 11 times that of the NPL (Table 2).

3.2. Water Chemistry Analysis

Among chemical analytes, ecoregion-specific differences existed between the blackwater and non-blackwater sites. Box plots do indicate that some sites we classified as non-blackwater did have a Color or DOC result that met our established study threshold, but to be classified as blackwater, sites had to meet thresholds for both analytes. In the CPL, NAP, and UMW, pH tended to be lower in blackwater sites than non-blackwater sites, with some sites in the CPL having notably low pH. Differences in pH between blackwater and non-blackwater in the NPL and TPL were negligible. Nutrients (Total N and Total P) were higher in blackwater sites of the NPL, and to a lesser degree in NAP. Conductivity was lower in blackwater sites than non-blackwater sites in the NAP and UMW. ANC was higher in NPL blackwater sites, lower for those in the UMW, and to a lesser degree also those in the CPL and NAP. Turbidity was not different between blackwater and non-blackwater sites for any ecoregion. Looking specifically at blackwater sites, boxplots of water chemistry variables showed that conductivity and ANC most strongly distinguished TPL and NPL sites from those in other ecoregions (Figure 2). Sites in the NPL and TPL also tended to have lower Color than those in the other ecoregions. Results for DOC did not vary greatly across ecoregions. Other parameters showed some variation among ecoregions but to a lesser degree.

For the Principal Components Analysis (PCA) of blackwater chemistry data, one blackwater site was missing some chemistry data, reducing the number of sites available for analysis from 310 to 309. Prior to analysis, we checked for significant correlations (Spearman $r > 0.80$) among analytes. ANC, pH, and conductivity were highly correlated, and because conductivity is more commonly measured by states and tribes, we excluded ANC and pH

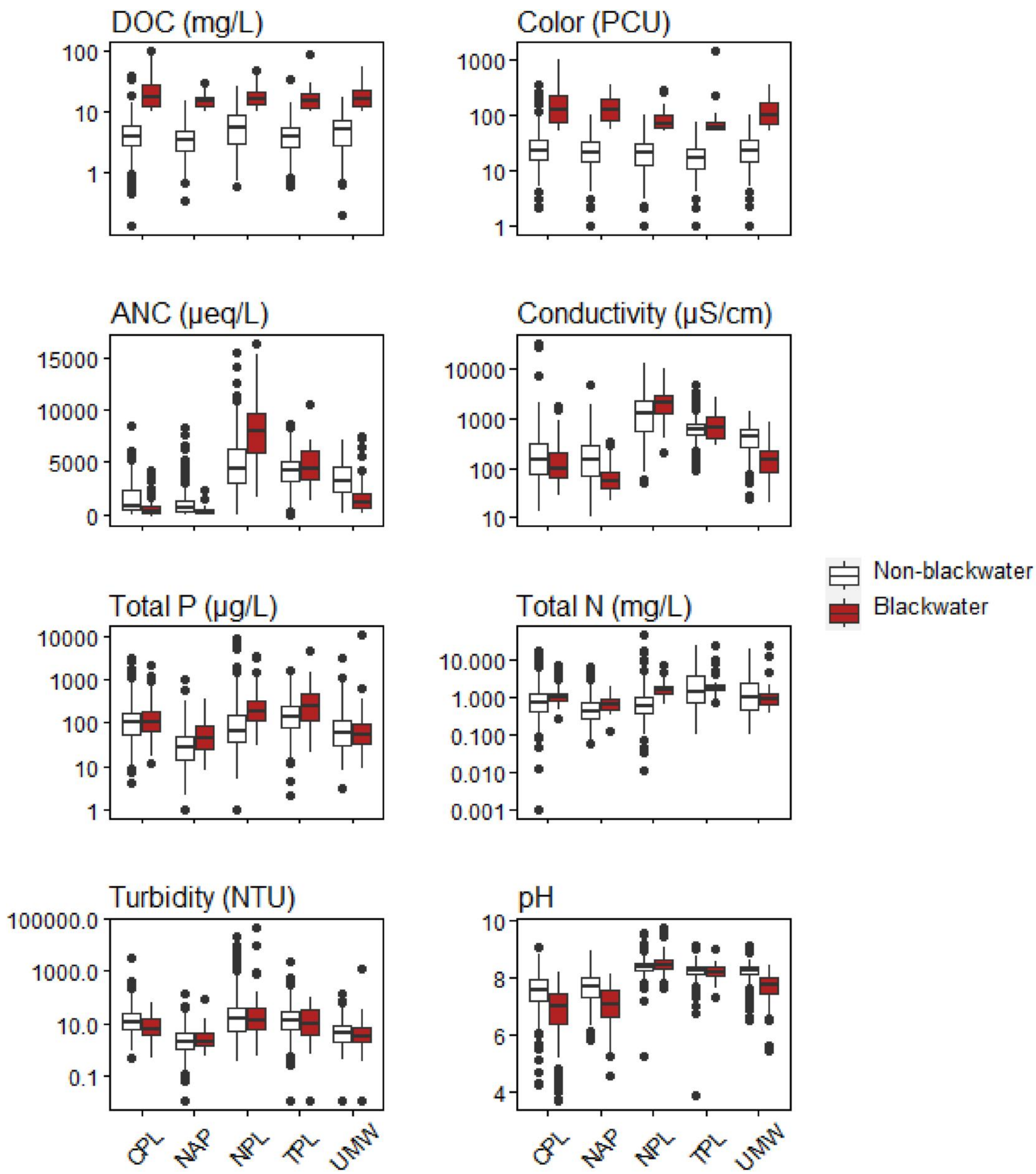

Figure 2. Box-and-whisker plots of water chemistry analytes comparing blackwater and non-blackwater sites by ecoregion. The upper and lower edges of the box (hinges) represent the 75th and 25th percentiles respectively. The horizontal line in the box represents the median. The lines extending from the box (whiskers) represent the furthest values within 1.5 times the box length (interquartile range) from the hinges. Points beyond the whiskers represent outliers. See Table 1 and Figure 1 for acronyms and Table S1 for number of sites. CPL-Coastal Plains, NAP-Northern Appalachians, NPL-Northern Plains, TPL-Temperate Plains, UMW-Upper Midwest.

from the set of variables used in the PCA. Importantly, we note that these correlations may not hold true in smaller scale studies and that there can be significant differences between acidic and alkaline high conductivity waters.

Collectively, the first two axes (PC1 and PC2) explained 75% of the variation (Figure 3). The NPL and TPL sites drive the PCA to some extent because they are so different from most other blackwater sites along PC1. The NAP sites are also clearly different and grouped along PC1. The other ecoregions overlapped considerably on PC1. There is also substantial overlap across ecoregions along PC2 meaning none of the water chemistry variables distinguished among ecoregions. The highest loadings along PC1 are turbidity (−0.47), conductivity (−0.45), Total P (−0.53), and Total N (−0.49). Color (−0.65) and DOC (−0.67) are the most important along PC2. PC2 represents the amount of carbon in the water (i.e., Color, DOC) while PC1 represents a water hardness axis

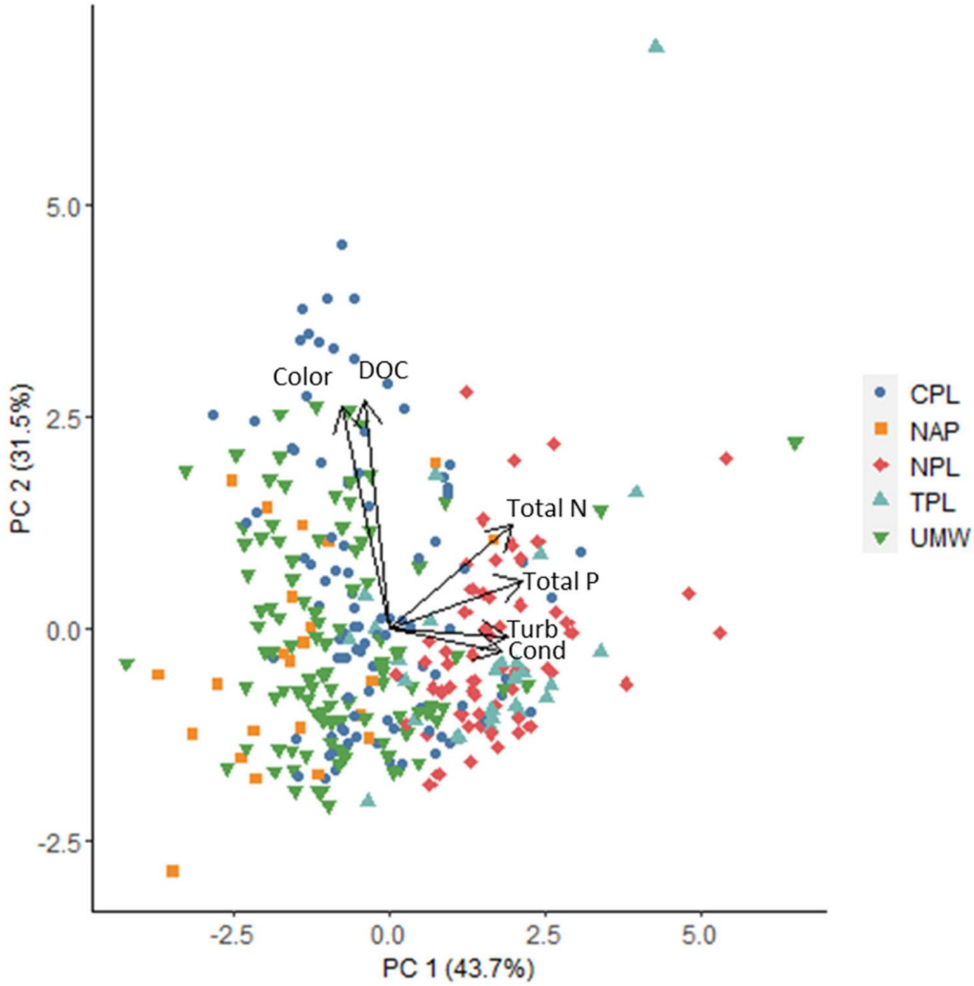

Figure 3. Principal Component Analysis of water chemistry variables. Blackwater sites only. See Table 1 and Figure 1 for acronyms and Table S1 for number of sites. PC1 represents a water hardness axis controlled by geology/soils. PC2 represents the amount of carbon in the water. CPL-Coastal Plains, NAP-Northern Appalachians, NPL-Northern Plains, TPL-Temperate Plains, UMW-Upper Midwest.

controlled by geology/soils. The ecoregions fall out along gradients that might be expected on PC1 and overlap strongly on the PC2 axis based on our definition of blackwater streams.

3.3. Physical Habitat Analysis

Several physical habitat measures were strongly correlated (Spearman $r > |0.80|$): (a) percent fines and mean substrate diameter, (b) mean wetted width × thalweg depth and mean wetted width, (c) mean thalweg depth and mean wetted width, and (d) mean thalweg depth and mean wetted width × thalweg depth. Because of these correlations, mean substrate diameter, mean wetted width × thalweg depth, and mean wetted width were dropped from PCA and the analysis of associations with blackwater conditions.

Figure 4 shows overlap between the physical habitat measures within blackwater and non-blackwater sites and no clear patterns of differences across ecoregions. Blackwater sites in the NPL and NAP did tend to have more fines (% Fines). Overall, these findings contradict expectations from prior studies which generally characterize blackwaters as being low-gradient and having a sandy bottom (e.g., Benke & Wallace, 2015; Colvin et al., 2020; Meyer, 1990). This lack of concordance with the literature likely results from sites in those studies being selected based on their unique nature whereas sites in our study were randomly selected from a pool of all possible sites in the CONUS, including smaller order streams.

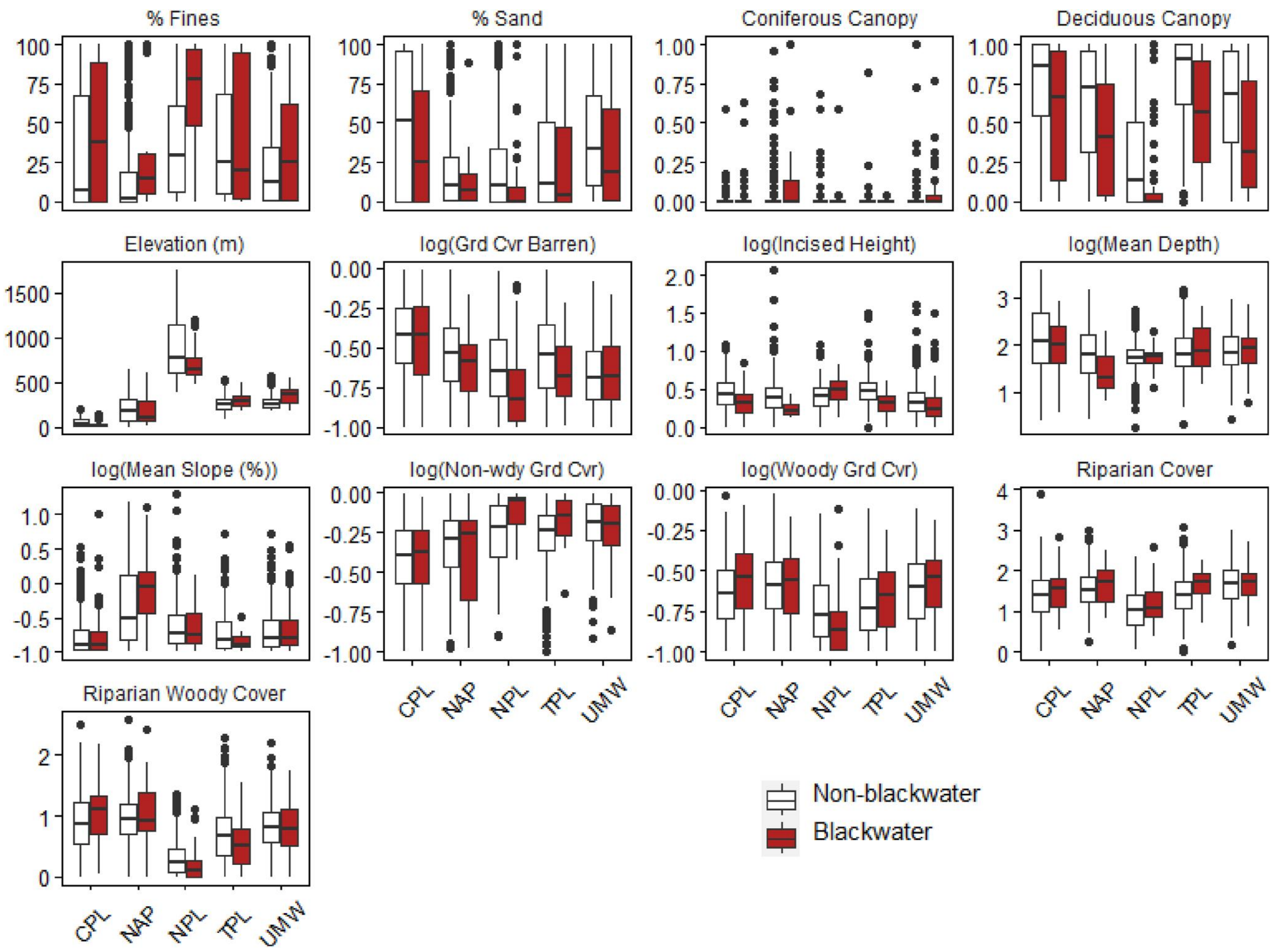

Figure 4. Box-and-whisker plots of physical habitat measures comparing blackwater and non-blackwater sites by ecoregion. See Table 1 and Figure 1 for acronyms and Table S1 for number of sites. CPL-Coastal Plains, NAP-Northern Appalachians, NPL-Northern Plains, TPL-Temperate Plains, UMW-Upper Midwest.

As expected, the PCA for physical habitat characteristics (Figure 5) did not explain much of the variation given that blackwater occurrence is inherently related to water chemistry; only 43% of the variation is explained by PC1 and PC2 for physical habitat compared to 75% for water chemistry. The NPL was distinct from other ecoregions, but other ecoregions did not strongly separate. On PC1, mean woody cover (Riparian Woody Cover, 0.45), elevation (ELEVATION, −0.37), and woody ground cover (Woody Grd Cvr, 0.36) were important in explaining that axis (i.e., highest loadings). On PC1, higher woody cover and woody ground cover were on one end, and higher elevation and substrate fines (% Fines, −0.32) on the other. PC2 was mostly driven by mean depth (Mean Depth, 0.55), percent slope (% Slope, −0.54), mean non-woody ground cover (Non-woody Grd Cvr, 0.33), and mean barren ground cover (Grd Cvr Barren, 0.26). In this case, higher slopes and more barren ground were on one end and greater depth and more non-woody ground cover on the other.

3.4. Wetland Analysis

Examination of the data set for the five ecoregions included in this study revealed that 59 sites were not in the catchments covered by StreamCat and therefore missing information critical to analysis. Of these, five were blackwater sites. This resulted in an available data set of 299 blackwater and 2028 non-blackwater sites for comparison. The lack of wetland habitat in general in drainages of the NPL sites made the comparison between blackwater and non-blackwater sites inappropriate. Thus, we focus on the remaining four ecoregions for the wetland analysis.

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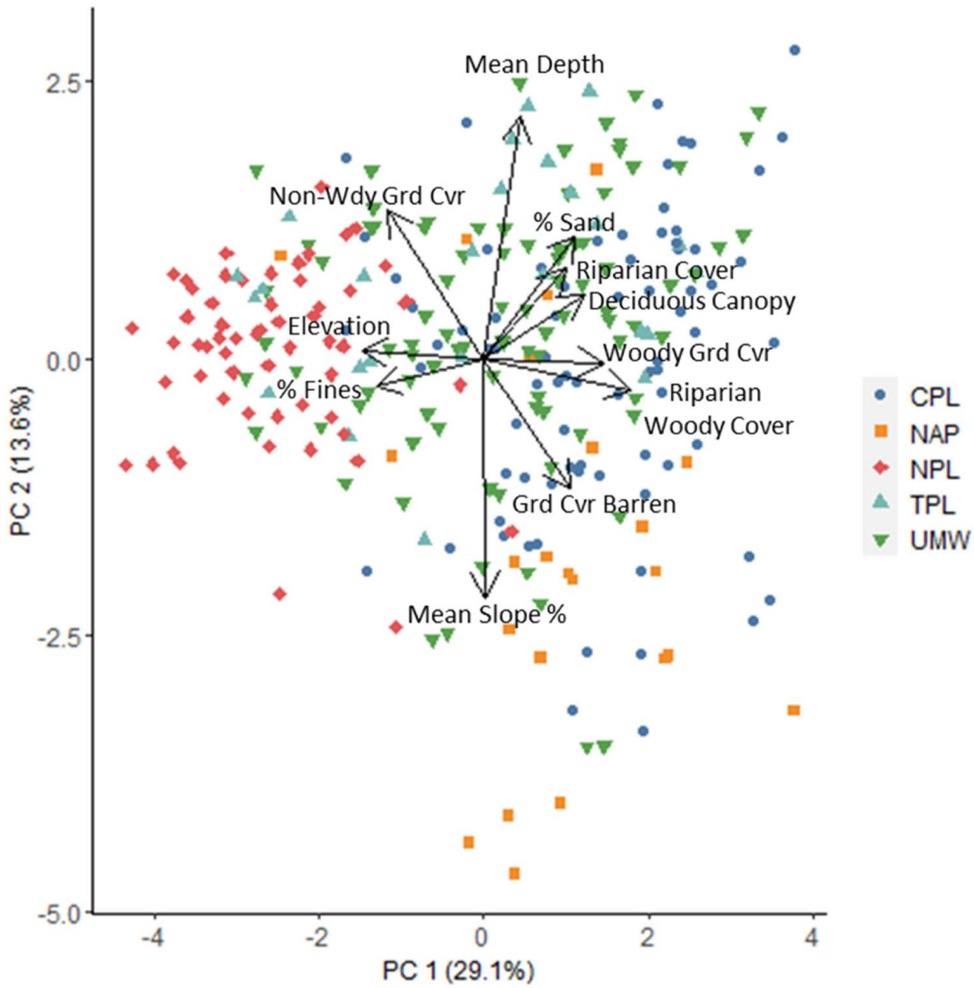

Figure 5. Principal Component Analysis of physical habitat variables with ecoregions identified. See Table 1 and Figure 1 for acronyms and Table S1 for number of sites. CPL-Coastal Plains, NAP-Northern Appalachians, NPL-Northern Plains, TPL-Temperate Plains, UMW-Upper Midwest.

For analysis of Euclidean distance to wetlands, 57 sites were missing information critical for our analysis, 12 of which were blackwater sites. Euclidean distance to wetlands was greater for non-blackwater sites in the CPL, NAP, TPL, and UMW although less pronounced in the CPL (Figure 6). Differences appeared negligible in the NPL, but again, the overall low frequency of wetlands in this ecoregion limits conclusions from these data. Regarding percent wetland in watersheds, across ecoregions, woody wetlands were more dominant than herbaceous wetlands. In the CPL, NAP, TPL, and UMW, drainages of blackwater sites had significantly greater woody wetland land cover than those for non-blackwater sites (Figure 7). The percentages of herbaceous wetland land cover in blackwater drainages were also greater in the CPL, TPL, and UMW. Unsurprisingly, the percentages of total wetland land cover (woody and herbaceous combined) were also greater in the contributing drainages of blackwater sites of the CPL, NAP, TPL, and UMW than in corresponding non-blackwater drainages.

4. Discussion

As stated earlier, sites identified as blackwater using our criteria were concentrated in five of the nine CONUS aggregated ecoregions with the distributional extent of many of the sites bounded by two major geomorphic features (Figure 1). Notably, there are blackwater sites in the Northern Plains Ecoregion (NPL) and the extreme western extent of the Coastal Plains Ecoregion (CPL) that are outside of the areas described in the previous paragraph. While the Fall Line is not present in the Western Gulf Coastal Plain, sites west of the Atlantic Fall Line have a similar topography as the Atlantic Coastal Plain (Daigle et al., 2006; Griffith et al., 2007). Those in the

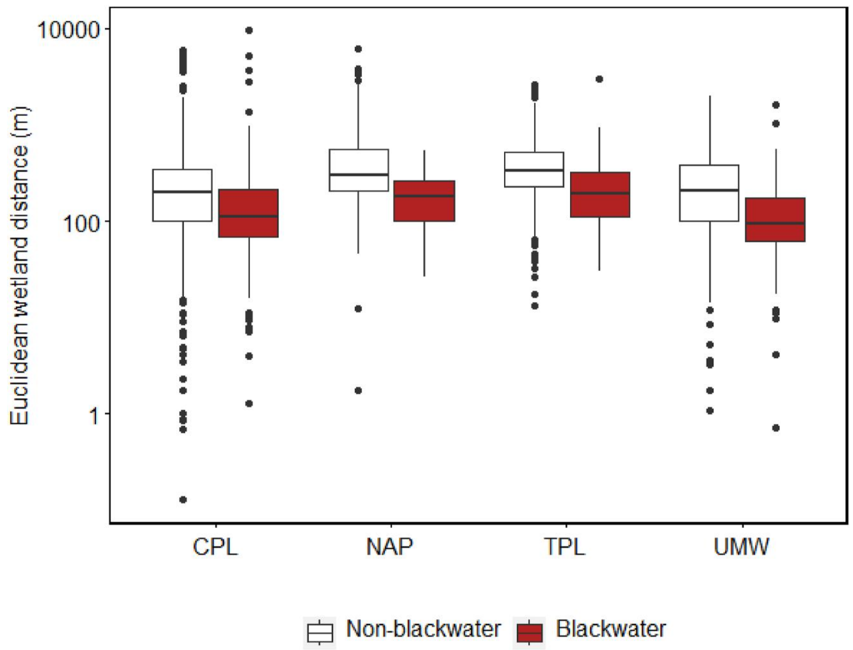

Figure 6. Box-and whisker plots of Euclidean distance (m) of blackwater and non-blackwater sites to a National Wetlands Inventory (NWI) wetland by ecoregion. See Table 1 and Figure 1 for acronyms and Table S1 for number of sites. CPL-Coastal Plains, NAP-Northern Appalachians, UMW-Upper Midwest.

NPL are concentrated in North Dakota southwest of the Missouri River, mostly in the Missouri Plateau Level IV ecoregion, an area not subjected to glaciation, and characterized as semiarid plains with occasional buttes and badlands (Bryce et al., 1998). Much of that area is underlain by lignite (Bluemle, 2011), which is a moisture-rich form of coal intermediate between peat and bituminous coal (Vorres, 2010). It is formed by the partial decomposition of peat deposited in Paleocene swamps 50–70 million years ago at relatively shallow depths (https://lignite.com/). These shallow depths resulted in the peat being exposed to relatively lower pressures and temperatures, preventing transformation to denser and drier forms of coal that are more carbon-rich (Schweinfurth, 2003; USEIA, 2021). Decomposition of organic material is limited in these deposit, such that remnants of woody fibers can often be observed (Mobley, 2001). Exposed lignite coal seams, either a natural occurrence or because of mining by early settlers (Oihus, 1983), can also serve as a source of tannins to surface waters. This is not suggested as the source of tannins for all sites in this region but is presented here as a likely explanation for many blackwater sites in the NPL.

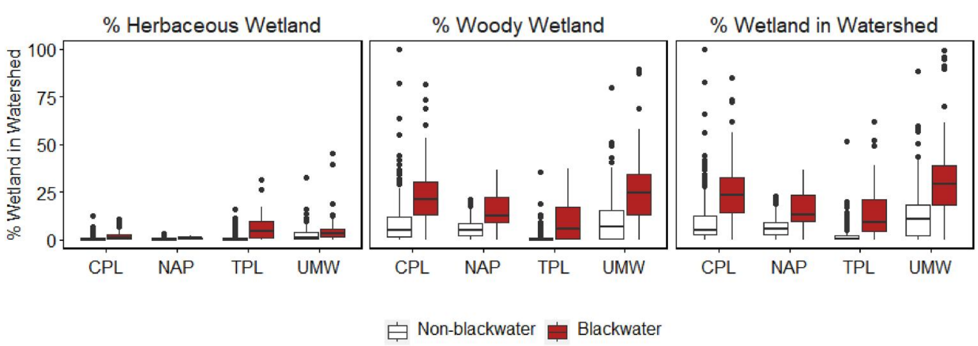

Figure 7. Box-and-whisker plots showing percent of herbaceous, woody, and total wetland in the upstream watersheds of blackwater and non-blackwater sites in the different ecoregions. See Table 1 and Figure 1 for acronyms and Table S1 for number of sites. CPL-Coastal Plains, NAP-Northern Appalachians, UMW-Upper Midwest.

10.1029/2023WR035529

As for blackwater sites located in Texas (TX), Louisiana (LA), and Arkansas (AR), many of these are within the floodplain of the Mississippi River, but were more than triple the distance from wetlands than sites in other states of the CPL (Euclidean distance of 308.2 km for TX + LA + AR vs. 94.3 km for CPL), suggesting an alternate source of tannins. Like those in the NPL, blackwater study sites in these states may also be related to lignite deposits. Although lignite deposits in this part of the country are not as vast as those in the NPL, they do exist in the area of blackwater sites in these states (Berge Exploration Inc, 1978; Schweinfurth, 2003). Further, North Dakota (ND) and TX combined account for 93% of U.S. lignite production (USDOE, 2021), which contributes to potential seam exposure in these areas.

Although NRSA sites that met our blackwater criteria in the WMT ecoregion of the CONUS were not identified, tannic systems do exist in that ecoregion although they are rare. An example is Henry's Fork in southeastern Idaho, which is a tributary of the Snake River. Additional examples are found near Yellowstone National Park (primarily in Wyoming) and include the Gibbon, Firehole, Madison, and Fall rivers (and some of their tributaries) along with Robinson Creek. Many, if not all, of these drain the southwest corner of Yellowstone Park and adjacent national forest land where wetlands are abundant. However, lower densities of tannic sites in the WMT decreases the likelihood of such sites being sampled as part of the NRSA. An additional challenge to sampling rare WMT blackwater systems is that the NRSA sampling season is set to coincide with summer low-flow, which may mask blackwater occurrence if tannins are being flushed into these streams during higher flow events. Lack of representation does not discount the potential existence of blackwater channels in the western US or in other areas, or their importance for inclusion in future research efforts.

Estimation of blackwater stream length across all five ecoregions included in this study was almost 172,000 km. This is comparable to the total perennial stream kilometers in the CPL, TPL, or UMW ecoregions, or the total stream length in the NPL, SPL, and XER ecoregions combined. This estimate should be considered conservative as it is based on samples taken during the summer low-flow survey season used by the NRSA survey. These estimations logically exclude, for example, blackwater rivers and streams that only flow as blackwater episodically during high-flow events (e.g., fill and spill wetlands) (Leibowitz, Mushet, & Newton, 2018). Beyond the relevance to estimates of stream length, seasonal blackwater flows are an important consideration in stream condition assessment as the water chemistry of the system when it is more characteristically blackwater (i.e., not at low-flow conditions) contributes to shaping the aquatic community year-round.

Comparison of water chemistry at blackwater sites among ecoregions showed that values for some water chemistry parameters in the NPL and the TPL differed from those of the UMW, the NAP, and CPL ecoregions (Figure 2). These differences are clearly visible in the water chemistry PCA that shows the extent that NPL and TPL sites drive the separation of ecoregions (Figure 3). These results may suggest that the sources of high Color and high DOC in these systems may be different from those in other ecoregions. Blackwater sites of the NAP are also clearly different and grouped together along PC1. Water chemistry in blackwater sites in the UMW and the CPL were remarkably similar (Figure 2), despite extensive geographic separation between these two ecoregions (Figure 1).

When examining physical habitat of different ecoregions, differences are apparent in the box-and-whisker plots and the PCA, but this is expected given that the ecoregions used were established to recognize areas with generally similar ecosystems (Omernik, 1987). However, the PCA does clarify that there are some ecoregional differences, and like the water chemistry, supports the argument that generalizations about blackwater rivers across ecoregions cannot be made. Within ecoregions, the physical habitat analysis did not show major differences between blackwater versus non-blackwater sites. The lack of concurrence between the patterns of percent fines and percent slope in Figure 5 seem to contradict this, but this is because the vectors for these two variables are responding on different PCA axes.

Multiple factors likely contribute to this general lack of differences across physical habitat measures within ecoregions. An important first consideration is that the water influencing the prevailing chemistry at a site classified as blackwater is, in most cases, contributed to the wetted channel by off-channel habitats (Benke & Wallace, 2015; Colvin et al., 2020; Flotemersch, 2023; Meyer, 1990; Smock & Gilinsky, 1992) and not the result of an autochthonous process within the wetted channel. The exception to tannin-rich water being primarily a product of off-channel habitats is where instream conditions contribute tannins to the stream (e.g., instream swamps; Todd et al., 2009). However, these instream situations do not typically occur to the extent that the system would be considered "blackwater." Once tannic water has entered the wetted channel,

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the water will retain its blackwater character until influenced by other factors altering the chemistry (e.g., dilution, geology).

Accepting that most sites identified as blackwater receive inputs from upstream off-channel habitats, it is reasonable to deduce that the majority of river and stream segments identified as blackwater in this study, where sites were randomly selected, are not adjacent to habitats serving as a source of tannic water. Therefore, when examining the physical habitat of these sites, and given that the forces controlling the broadscale environment at a river or stream (e.g., climate, topography, geology, landcover) should all be relatively consistent in a given area (in this case, ecoregion; Grant et al., 2003; Snelder & Biggs, 2002), a finding that there is not much difference between the physical habitat of blackwater and non-blackwater sites seems reasonable. If we were to limit our comparison of blackwater and non-blackwater sites to only include blackwater sites adjacent to sources of tannins (i.e., wetlands), results would likely differ as the physical habitat of the geomorphic setting of wetlands would logically be less variable than physical habitat measures derived from a pool of all blackwater sites and potentially different from that of non-blackwater sites.

The relationship of wetlands to blackwaters is well documented (e.g., Benke et al., 1984; Smock & Gilinsky, 1992). Our study is the first to demonstrate this relationship in multiple regions of the CONUS. Blackwater sites were closer (Euclidian distance) to the nearest wetland than non-blackwater sites in all ecoregions included in the study, with the exception of the NPL, where the distance of all sites (i.e., blackwater and non-blackwater) to the nearest wetland was greater than in the other ecoregions. This finding supports our suggestion of alternative sources of tannins in blackwaters of that ecoregion (NPL). Examination of the percentage of wetlands in watersheds showed that blackwaters had a higher percentage of wetland land cover compared to non-blackwater systems. More specifically, the percentage of woody wetlands was greater in blackwater stream drainages in all ecoregions, and the percentage of herbaceous wetland was higher in most.

5. Conclusion

Blackwaters are a unique aquatic habitat with a substantial presence in multiple ecoregions of the CONUS. Using our study criteria, blackwaters constitute about 15% of the stream length across the five ecoregions that were the focus of our study. The water chemistry of blackwater rivers and streams in some ecoregions can be substantially different from that of non-blackwaters. Such compositional differences could contribute to inaccurate water quality condition assessments of blackwater rivers and streams because the basis of river and stream assessments reflect the natural conditions observed in non-blackwater systems. Analysis of physical habitat data did not show dramatic differences between blackwater and non-blackwater sites within or across ecoregions but did challenge the common perception that blackwaters are universally low gradient and consist of sandy substrates. As expected, wetlands were shown to be an important factor contributing to the presence of blackwater rivers and streams, although some wetlands were buried rather than contemporary surface wetlands. Taken collectively, the water chemistry and physical habitat data strongly support the conclusion that systems considered blackwater possess varying characteristics, both within ecoregions of United States, and that a classification framework that accounts for this variability is essential to their protection.

Given these findings, we propose the following research to support improved understanding and protection of blackwater rivers and streams of the CONUS.

(1) Development of an operational definition of blackwater rivers and streams that is broad enough to cover the systems described in this paper. This working definition may include sites that are ultimately excluded in more refined definitions of "blackwater," yet represent river and stream habitats in need of further study.
(2) Development of a classification system that recognizes the myriad of generalized types of blackwater systems (e.g., low vs. high pH). The development of this classification system should be structured so that it at once increases the awareness and understanding of blackwaters and provides water resource managers the capability to accurately identify them.
(3) Identification of least-disturbed reference conditions for different types of blackwater rivers and streams identified in the classification system. Defining these reference conditions will be of scientific value but also be of high utility to resource managers tasked with the protection of blackwater resources.

(4) Identification of stressors and their associated dynamics that negatively impact blackwater rivers and streams. For example, nutrient levels impacting some types of blackwater systems may be much lower than for non-blackwater systems. Understanding these dynamics are necessary to support protection efforts.
(5) Development of data-driven, consistent, and repeatable assessment methods, including development of targets, protective of unique conditions in blackwater rivers and streams, where needed. For example, for a given type of blackwater river or stream, what pH levels might be expected?
(6) Development of methods to distinguish natural from human-disturbed systems with high DOC. For example, in contrast to systems with naturally high DOC, a system may have high DOC due to inadequate treatment of wastewater, or leachate from landfills (Teng et al., 2021).

Addressing these research needs will contribute significantly to the protection of these unique aquatic habitats across the United States and support other global efforts with the same objectives.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The site, chemistry, and habitat data sets and metadata on which this article is based are available in USEPA-NARS-DATA (2023). The StreamCat wetland data used in this article are available at USEPA-NARS-STREAMCAT (2023).

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